

Oral communication abstracts

OC01: PLENARY: TOP ABSTRACTS 2026

OC01.01

ASTRiD-M study: a prospective multicenter evaluation of the predictive value of the ‘amniotic sac sign’ in the diagnosis of early miscarriage

S.A. Solangon⁴, M. Morje¹, S. Platts³, F. Taki², G. Premetis⁵, K. Vogt⁶, S. Syed⁷, J.A. Ross³, D. Jurkovic^{4,1}

¹University College London Hospitals NHS Foundation Trust, London, United Kingdom; ²St Mary's Hospital, London, United Kingdom; ³King's College Hospital, London, United Kingdom; ⁴Institute for Women's Health, University College London, London, United Kingdom; ⁵Royal Free London NHS Foundation Trust, London, United Kingdom; ⁶Whittington Hospital, London, United Kingdom; ⁷Hillingdon Hospitals NHS Foundation Trust, Uxbridge, United Kingdom

Objectives: A large single-center study showed that the presence of an amniotic sac without a live embryo with CRL <7mm on ultrasound scan was accurate in the detection of miscarriage at the initial visit. Our aim was to externally validate this ‘amniotic sac sign’ by reassessing its specificity for miscarriage and assessing the potential reduction in follow-up scans for pregnancies of uncertain viability (PUV).

Methods: This was a prospective multicenter cohort study of pregnant women less than 14 weeks of gestation, who attended an early pregnancy unit between January and August 2023. Outcomes included diagnosis by the first-trimester dating scan, number of follow-up visits and misdiagnosis of the ‘amniotic sac sign’.

Results: Our study population included 1865 women over five centers. A conclusive diagnosis was reached on initial scan in 1053 (56.5%) cases. 254/267 (95.1%) PUL completed follow-up, totalling 186 additional scans to reach a final diagnosis. 456/545 (83.7%) PUV completed follow-up, totalling 481 scans to reach a final diagnosis. 190 (34.9%) PUV had live pregnancies, 265 (48.6%) had a miscarriage and 1 (0.2%) had an ectopic pregnancy. 77 (16.9%) PUV had an amniotic sac without a live embryo with CRL <7mm. The ‘amniotic sac sign’ had a sensitivity of 29.1% (95%CI 23.6 to 34.93%), specificity of 100% (95%CI 98.1 to 100%), PPV of 100% (95%CI 95.32 to 100%), NPV of 76.7% (95%CI 75.3 to 78.0%) and accuracy of 78.7% (95%CI 74.7 to 82.4%) for the detection of miscarriage. By using the amniotic sac sign to diagnose miscarriage, the number of follow-up scans for PUV would be reduced by 16.2% (78/481), accounting for 11.7% of all follow-up scans for diagnostic uncertainty.

Conclusions: Our multicenter study has shown that the ‘amniotic sac sign’ has 100% specificity for the diagnosis of miscarriage. Application of this sign could reduce follow-up scans for pregnancies of uncertain viability by 16%, reducing patient anxiety and delivering system-level efficiencies.

OC01.02

***Ultrasound risk stratification of adnexal masses in pregnancy: results from the pregnancy-IOTA study**

N. Cooper^{9,1}, A. Ledger², B. Van Calster², A.C. Testa³, W. Froyman², S. Saso^{1,9}, C. Taliento¹¹, S. Guerra², L. Wong⁵, D. Djokovic⁶, F. Sorbi¹⁰, S. Del Forno⁷, L. Valentin⁸, D. Timmerman^{2,4}, T. Bourne^{9,2}

¹Imperial College Healthcare NHS Trust, London, United Kingdom; ²Katholieke Universiteit Leuven, Leuven, Belgium; ³Obstetrics and Gynecology Clinic, University Catholic of Sacred Heart, Rome, Italy; ⁴University Hospital KULeuven, Leuven, Belgium; ⁵Obstetrics and Gynecology, Monash University, Clayton, VIC, Australia; ⁶Obstetrics and Gynecology, Hospital CUF Descobertas, Lisbon, Portugal; ⁷Gynecology and Obstetrics, Sant'Orsola-Malpighi Hospital, University of Bologna, Bologna, Italy; ⁸Skåne University Hospital, Malmö, Sweden; ⁹Imperial College London, London, United Kingdom; ¹⁰Università degli Studi di Firenze, Florence, Italy; ¹¹Università degli Studi di Ferrara Facoltà di Medicina Farmacia e Prevenzione, Ferrara, Italy

Objectives: To evaluate the performance of the IOTA Benign Descriptors (BDs) and ADNEX model for the risk stratification of adnexal masses in pregnancy.

Methods: This interim analysis includes participants from the prospective, international multicenter pregnancy (P)-IOTA study. The BDs, ADNEX model and two-step were prospectively applied to ovarian masses diagnosed in pregnancy at any gestation. Three study arms were included: arms 1 and 2 underwent a single scan during pregnancy with postpartum follow-up, while arm 3 had longitudinal assessment at predefined gestational timepoints. Diagnostic performance of the ADNEX model and two-step strategy was assessed using the area under the curve (AUC), sensitivity, specificity, positive predictive value (PPV) and negative predictive value (NPV). CA-125 was included in the ADNEX model as per clinical practice.

Results: In total, 541 participants across 21 centers are recruited to date. This analysis includes 281 pregnant patients from 17 centers with an adnexal mass diagnosed during pregnancy. Overall, 188/281 (66.9%) tumors were benign, 28/281 (10.0%) malignant and 56/281 (19.9%) had uncertain outcome. The remaining 9 had outcomes recorded outside of the study protocol timepoints.

72 patients (72/281, 25.6%) underwent surgery, including 54/72 (75.0%) during pregnancy, of which 27/54 (50.0%) had the mass removed at Cesarean section. Three patients (3/72, 4.2%) were planned for surgery after pregnancy due to suspicion of malignancy.

The BDs were applicable in 122/281 tumors (43.4%), and all were benign (122/122, 100%). The ADNEX model achieved an AUC of 0.89 (95% CI 0.81-0.94), as did the two-step strategy (AUC 0.89, 95% CI 0.81-0.94).

Conclusions: The ADNEX model and two-step strategy showed high diagnostic performance for classifying adnexal masses in pregnancy. The BDs were applicable in a significant proportion of tumors and identified only benign lesions, supporting their use to guide safe conservative management of adnexal masses during pregnancy.

*This presentation is eligible for the Top Abstract and Young Investigator awards (to be presented in the closing plenary).